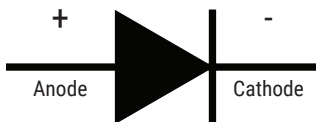


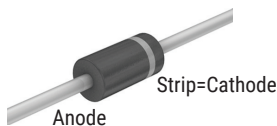
DIODES AND TRANSISTORS

Diode Basics

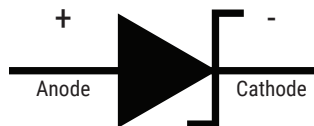
Junction Diodes and Rectifiers – two-element devices containing an anode and a cathode that provide current flow of electricity in one direction. Applications include: rectifying AC voltage for conversion to DC power supplies, used to prevent shorting when applying power, and clamping reverse current spikes on motors.



Diode schematic symbol



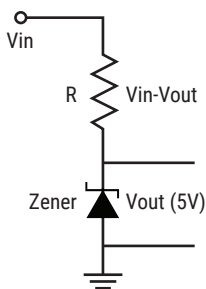
Axial Lead DO-41 Package



Zener Diode schematic symbol

Zener Diodes

Zener diodes are normally used as a voltage reference or voltage regulator when *reversed* bias to meet its breakdown voltage for conducting electricity in the opposite direction.



Junction Diodes

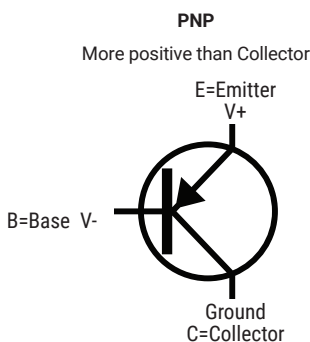
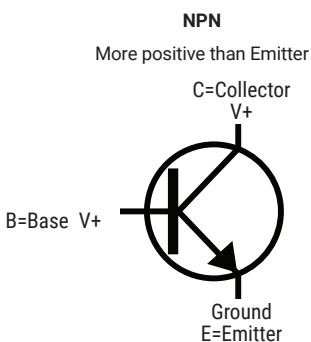
Example part number:
1N4001GDICT-ND

Zener Diodes

Example part number:
1N5232B-ND

Transistor Basics

A transistor is an electronic component that can act as a switch or an amplifier. There are two main types of transistors: voltage-enabled Field Effect Transistors or “FETs” and current-driven Bipolar Junction Transistors or “BJTs”. For the NPN BJT below, a small amount of current at the Base leads to a larger current flow from Collector to Emitter.



Reviewing the above schematic symbols, you should note that to turn on an NPN type transistor you will need to use positive current on the Base lead. To turn on PNP transistors you will need to use negative current or at ground level on the Base lead. Typically, NPN should have the load on Collector side which is controlled by positive current flowing to the Base which then switches the Emitter to ground allowing current to flow from Collector to Emitter is called sourcing the load. For PNP type transistors, the Emitter is on the positive side of the supply voltage and the load is on the Collector side but sinking it to ground via negative current going to the Base (listed as V-, Negative Voltage or at Ground).

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